R13

Code No: 118EQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
SEMANTIC WEB AND SOCIAL NETWORKS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks

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1. How does World Wide Web evolve into a resource with intelligent capabilities? Explain with suitable scenarios. [15]

2.a) Compare and contrast information revolution with agriculture revolution.
b) Explain how innovations in adaptive software may offer surprising capabilities for Web software agents. [7+8]

3.a) "XML is not a replacement, rather a complementary technology to HTML." Support this statement with suitable scenarios.
b) Illustrate the difference between RDF and RDFS layer.
c) How to introduce constraints on properties of RDF? [5+5+5]

4. Describe ontology with the domain of multi-specialty hospital. Use inference engine to validate your ontology and check for inconsistencies. [15]

5.a) What type of rules are preferred in logic where information is unavailable? How to handle conflicts between these rules? Explain with an example.
b) When does an inference engine use backward chaining instead of forward chaining? Give illustrations. [7+8]

6.a) What challenges are being faced by semantic based search engines?
b) How the OWL-S ontology-based Web services can be dynamically invoked by other services on the Web? Explain. [7+8]

7.a) Explain how is social network analysis a different approach to social phenomena.
b) How can a sociogram be constructed for an electronic discussion forum? Give illustrations. [7+8]

8.a) The biggest technical challenge in social network mining is the disambiguation of person names. Discuss this challenge and mechanism to handle it.
b) Write note on building web applications with social network features. [7+8]

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